

INTEGRALS INDEFINITE

1.	Find : $\int \frac{1}{\sqrt{x}(\sqrt{x} + 1)(\sqrt{x} + 2)} dx$
2.	Find : $\int \frac{1}{\cos(x - a) \cos(x - b)} dx$
3.	If $\frac{d}{dx} f(x) = 2x + \frac{3}{x}$ and $f(1) = 1$, then $f(x)$ is (A) $x^2 + 3 \log x + 1$ (B) $x^2 + 3 \log x$ (C) $2 - \frac{3}{x^2}$ (D) $x^2 + 3 \log x - 4$
4.	Find $\int \frac{\sin^{-1} x}{(1 - x^2)^{3/2}} dx.$
5.	Find $\int e^x \left(\frac{1 - \sin x}{1 - \cos x} \right) dx$
6.	$\int 2^{x+2} dx$ is equal to : (a) $2^{x+2} + C$ (b) $2^{x+2} \log 2 + C$ (c) $\frac{2^{x+2}}{\log 2} + C$ (d) $2 \cdot \frac{2^x}{\log 2} + C$
7.	Find : $\int \frac{dx}{\cos x \sqrt{\cos 2x}}$

8.	$\int e^{-x} \left(\frac{x+1}{x^2} \right) dx$ is equal to : (a) $\frac{e^{-x}}{x} + C$ (b) $\frac{e^x}{x} + C$ (c) $\frac{e^x}{x^2} + C$ (d) $-\frac{e^{-x}}{x} + C$
9.	Find : $\int \frac{\cos x}{\sin 3x} dx$
10.	Find : $\int x^2 \log (x^2 + 1) dx$
11.	$\int e^{5 \log x} dx$ is equal to : (a) $\frac{x^5}{5} + C$ (b) $\frac{x^6}{6} + C$ (c) $5x^4 + C$ (d) $6x^5 + C$
12.	Find : $\int \frac{x^2 + x + 1}{(x + 1)^2 (x + 2)} dx$
13.	Find : $\int \frac{e^x}{\sqrt{5 - 4e^x - e^{2x}}} dx$

14.	$\int \frac{\sec x}{\sec x - \tan x} dx$ equals (a) $\sec x - \tan x + c$ (c) $\tan x - \sec x + c$ (b) $\sec x + \tan x + c$ (d) $-(\sec x + \tan x) + c$
15.	Find : $\int e^{x^2} (x^5 + 2x^3) dx$
16.	$\int \frac{x-3}{(x-1)^3} e^x dx$ is equal to : (A) $\frac{2e^x}{(x-1)^3} + C$ (C) $\frac{e^x}{(x-1)} + C$ (B) $\frac{-2e^x}{(x-1)^2} + C$ (D) $\frac{e^x}{(x-1)^2} + C$
17.	Find : $\int \frac{x^3 - 1}{x^3 - x} dx$
18.	Find : $\int \frac{5x - 3}{\sqrt{1 + 4x - 2x^2}} dx$
19.	$\int \frac{\cos 2x}{\sin^2 x \cdot \cos^2 x} dx$ is equal to (a) $\tan x - \cot x + C$ (c) $\cot x + \tan x + C$ (b) $-\cot x - \tan x + C$ (d) $\tan x - \cot x - C$

20.	Find : $\int \frac{\cos \theta}{\sqrt{3 - 3 \sin \theta - \cos^2 \theta}} d\theta$
21.	Find : $\int \frac{2x^2 + 1}{x^2(x^2 + 4)} dx$
22.	Find : $\int x \sqrt{1 + 2x} dx$
23.	Find : $\int \frac{x^2}{(x^2 + 4)(x^2 + 9)} dx$
24.	Find : $\int x^2 \cdot \sin^{-1}(x^{3/2}) dx$
25.	Find : $\int \frac{1}{x [(\log x)^2 - 3 \log x - 4]} dx$

26.	Given $\frac{d}{dx} F(x) = \frac{1}{\sqrt{2x-x^2}}$ and $F(1) = 0$, find $F(x)$.
27.	Find : $\int \frac{\sqrt{x}}{(x+1)(x-1)} dx$
28.	Find : $\int \frac{2+\sin 2x}{1+\cos 2x} e^x dx$
29.	Find : $\int \frac{1}{x(x^2-1)} dx$.
30.	Find : $\int e^x \left[\frac{1}{(1+x^2)^{\frac{3}{2}}} + \frac{x}{\sqrt{1+x^2}} \right] dx$
31.	Find : $\int \frac{3x+5}{\sqrt{x^2+2x+4}} dx$

